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(57) Abstract :
The present invention relates to a secure communication system for wireless communication networks integrating Quantum Elliptic Curve Cryptography (QECC), Quantum Key Sharing (QKS), and trust-based routing. The system comprises evaluating neighbouring nodes based on packet forwarding behaviour, message integrity, delivery consistency, historical trust values, and recommendations from neighbouring nodes to generate a cumulative trust score. Nodes having a trust score below a predefined threshold are identified as malicious and excluded from secure route selection. QECC provides cryptographic protection, while QKS facilitates secure key establishment between trusted communicating nodes for encrypted data transmission. A network monitoring module continuously evaluates node behaviour and communication parameters, including packet delivery ratio, throughput, error rate, and energy consumption, to facilitate dynamic route updating. The integrated framework enhances confidentiality, authentication, privacy, and resistance to routing attacks while reducing communication overhead and energy consumption. The system applies to wireless sensor networks, mobile ad hoc networks, IoT, WBANs, smart healthcare, smart grid, and RFID-based communication systems.

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